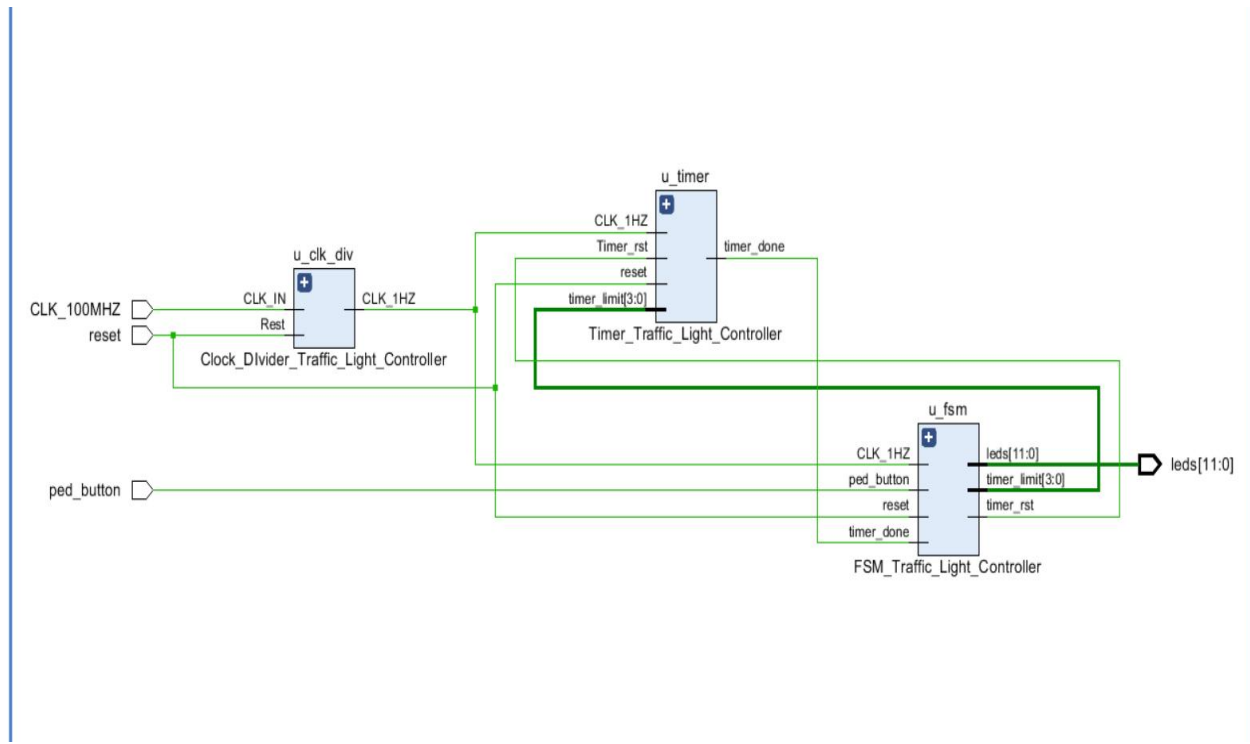
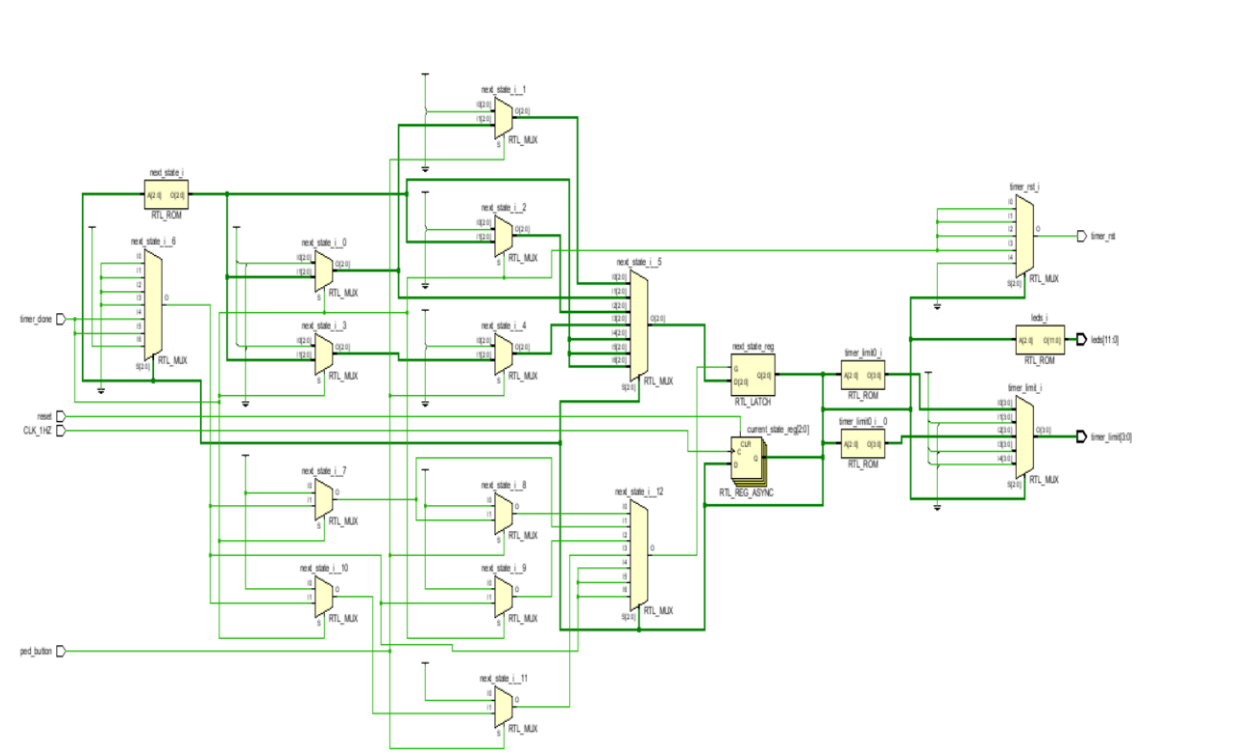


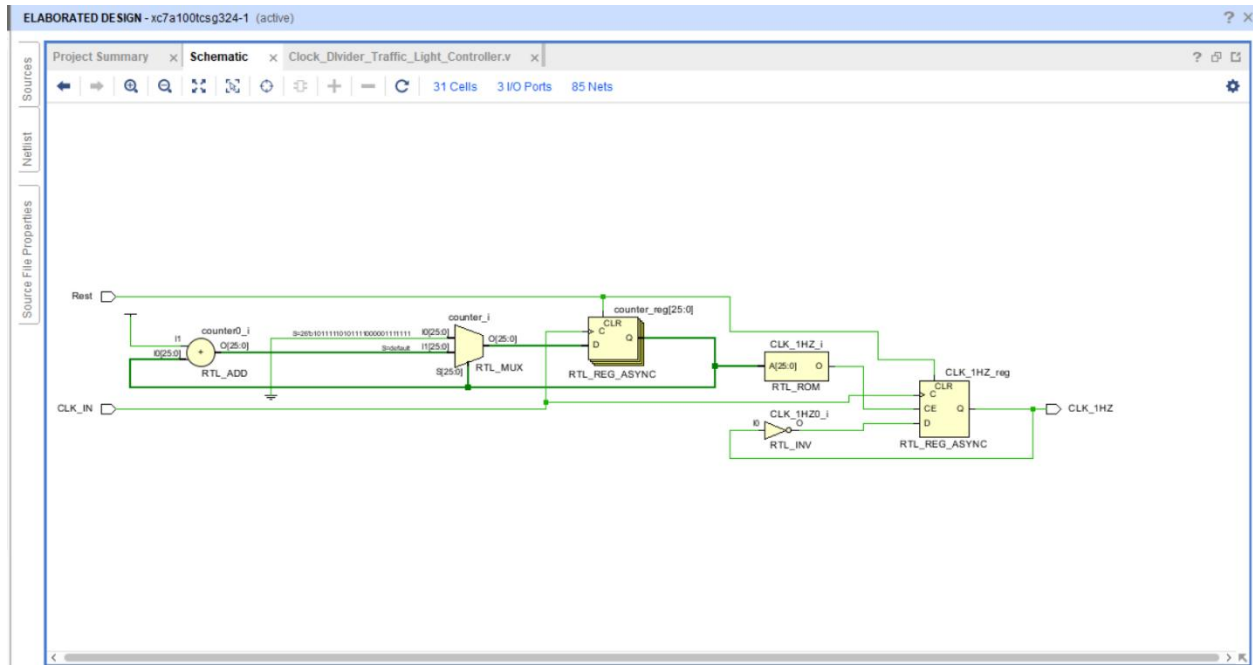
Block Diagram



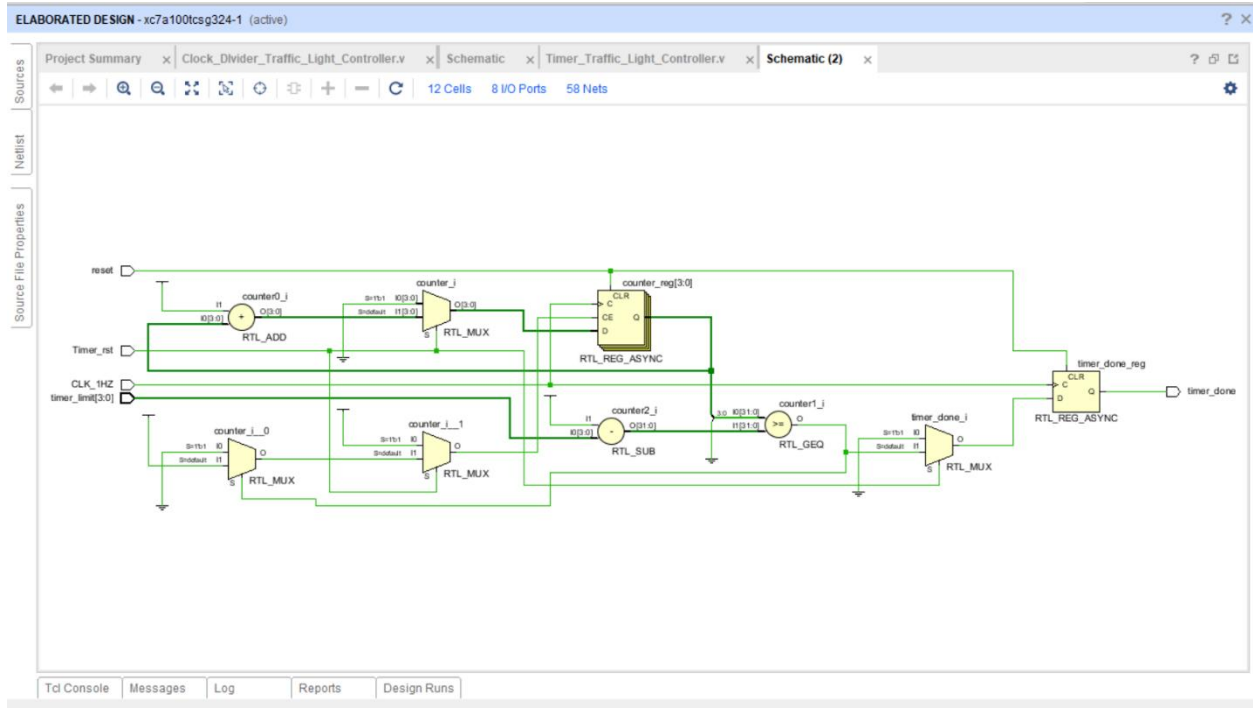
fsm



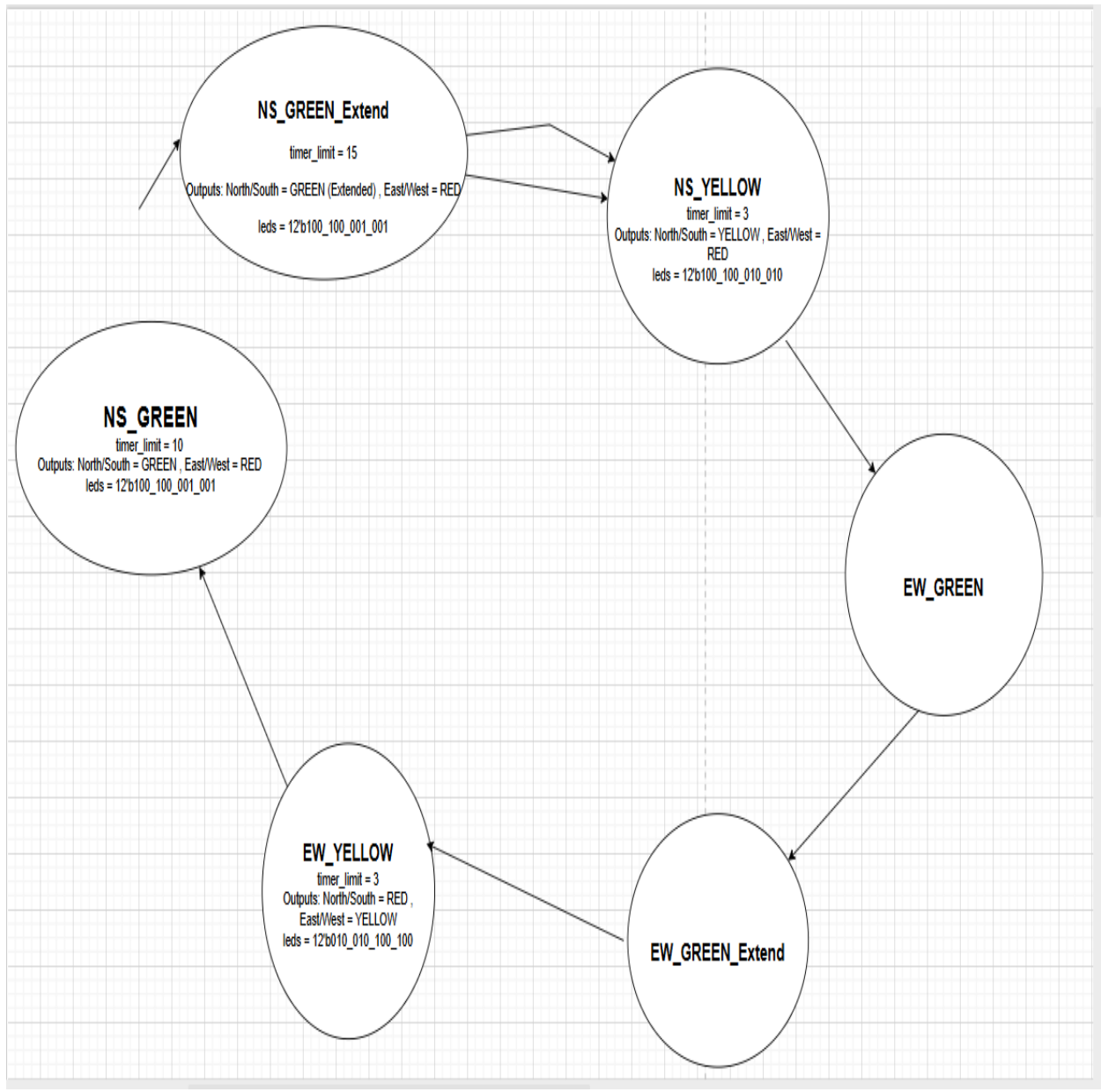
Clock divider



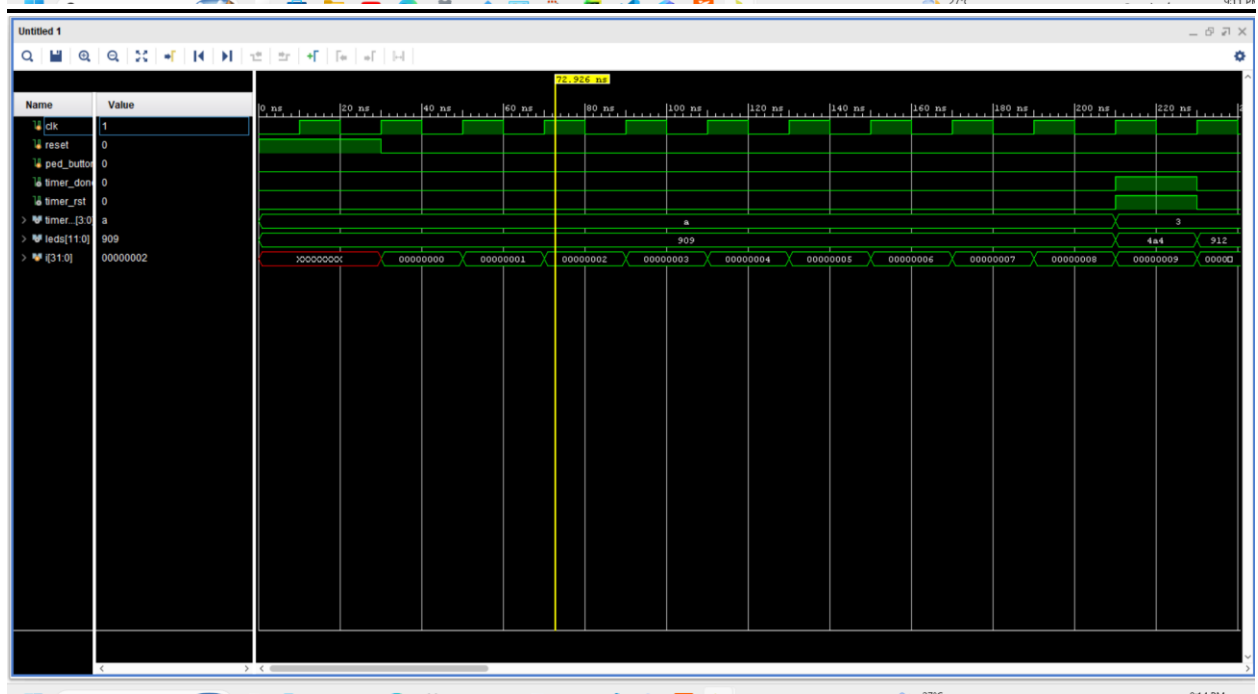
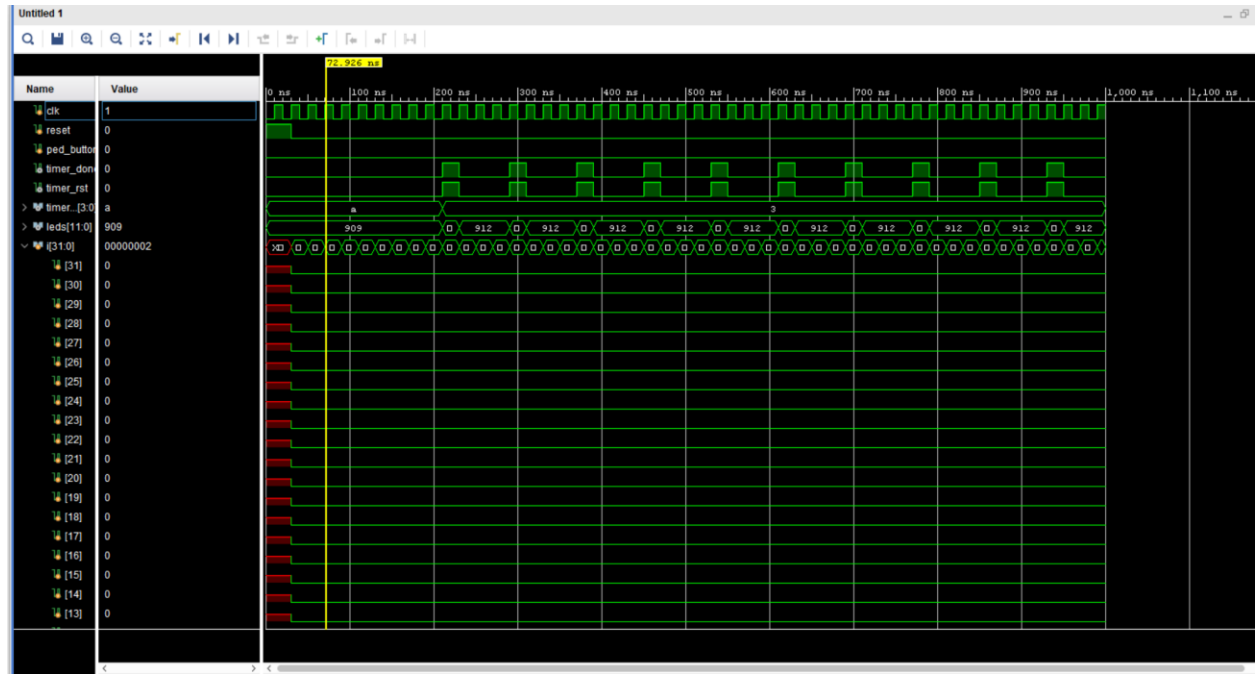
Timer

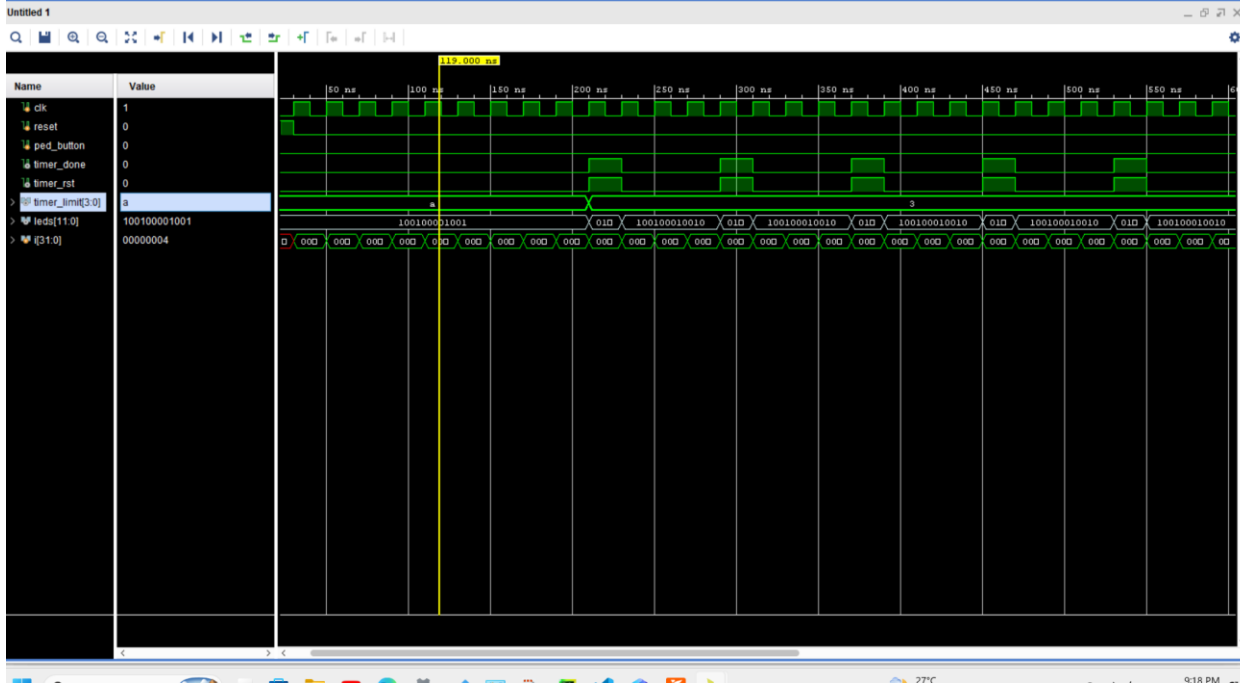


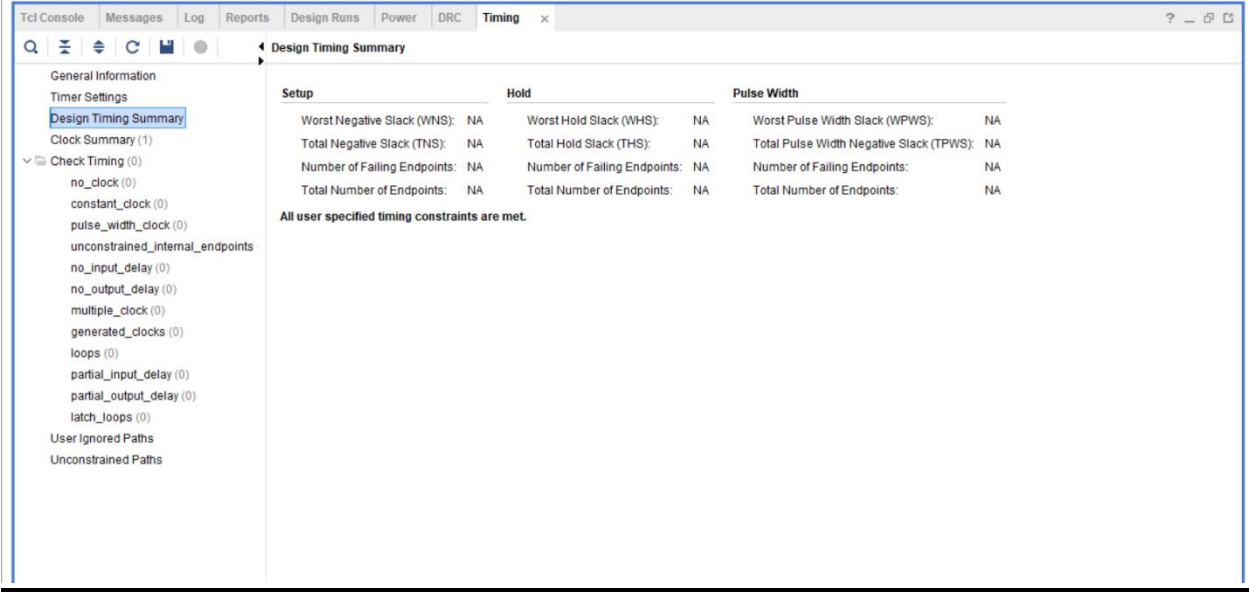
FSM State Diagram



Timing Diagram







Pin Mapping Table

<u>Description</u>		

Simulation Explanation

The functional behavior of the Top Traffic Light Controller was fully verified via behavioral simulation using a comprehensive testbench. The simulation waveform successfully demonstrates the following key engineering outcomes:

1. **Correct Sequential Flow:** The design transitions exactly as defined by the Finite State Machine (FSM) parameters. It accurately maintains the 10-second Green phases, 3-second Yellow phases, and implements the 5-second pedestrian extension (extending the green phase to 15 seconds) whenever the 'ped_button' is asserted.
2. **Safety and Mutually Exclusive Design:** The simulation results mathematically and logically prove the absence of any conflicting phases. Specifically, there is absolutely NO overlapping "Green" state in perpendicular directions. While one direction (North-South) is Green or Yellow, the perpendicular direction (East-West) is strictly forced to Red, satisfying the safety critical requirement of the system.
3. **Proper Timer Synchronization:** The internal timer counts sequentially and asserts 'timer_done' right at the specific boundaries, resetting safely at every state change without creating any timing glitches or undefined states.

FPGA Test Results

The digital design was successfully synthesized, implemented, and verified using Xilinx Vivado Design Suite for the targeted Artix-7 FPGA architecture. The engineering reports confirm the following status:

1. **Synthesis and Implementation Status:** Both the Synthesis and Implementation phases were completed successfully with ZERO errors and ZERO critical warnings. The tool successfully mapped the modular RTL logic into the physical hardware primitives of the FPGA chip.
2. **Hardware Resource Utilization:** The hardware utilization report shows optimal area efficiency. The entire design utilizes exactly 12 Bonded Input/Output Blocks (IOBs), corresponding directly to the mapped physical pins on the board (12 LEDs for the traffic intersections, 1 clock input, and 2 push buttons).

3. Timing Constraints Summary: According to the official Vivado Timing Summary report, "All user specified timing constraints are met." The design achieved complete structural timing closure with no setup or hold time violations, confirming that the physical hardware will run reliably and stably under the 100 MHz on-board system clock.

4. Bitstream Generation: The final Bitstream file (.bit) was successfully written and generated without errors. It is ready to be programmed directly onto the hardware board for real-time testing.

The image displays two screenshots of the Vivado IDE interface. The top screenshot shows the 'Design Timing Summary' report, which confirms that all user-specified timing constraints are met. The bottom screenshot shows the 'Utilization' report, highlighting the hierarchy of the design, including the 'Bonded IOB' and 'Top_Traffic_Light_Control...' block.

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): NA	Worst Hold Slack (WHS): NA	Worst Pulse Width Slack (WPWS): NA
Total Negative Slack (TNS): NA	Total Hold Slack (THS): NA	Total Pulse Width Negative Slack (TPWS): NA
Number of Failing Endpoints: NA	Number of Failing Endpoints: NA	Number of Failing Endpoints: NA
Total Number of Endpoints: NA	Total Number of Endpoints: NA	Total Number of Endpoints: NA

All user specified timing constraints are met.

Utilization

Name	Bonded IOB (210)
Top_Traffic_Light_Control...	12

Report						Options		Modified	Size
impl_1_place_report_utilization_0								7/11/26 8:52 PM	7.4 KB
impl_1_place_report_control_sets_0								7/11/26 8:52 PM	2.7 KB
impl_1_place_report_incremental_reuse_0									
impl_1_place_report_incremental_reuse_1									
impl_1_place_report_timing_summary_0									
Post-Place Power Opt Design (post_place_power_opt_design)									
impl_1_post_place_power_opt_report_timing_summary_0									
Post-Place Phys Opt Design (phys_opt_design)									
impl_1_phys_opt_report_timing_summary_0									
Route Design (route_design)									
impl_1_route_report_drc_0								7/11/26 8:52 PM	2.5 KB
impl_1_route_report_methodology_0								7/11/26 8:52 PM	1.5 KB
impl_1_route_report_power_0								7/11/26 8:52 PM	6.2 KB
impl_1_route_report_route_status_0								7/11/26 8:52 PM	0.6 KB
impl_1_route_report_timing_summary_0								7/11/26 8:52 PM	7.6 KB
impl_1_route_report_incremental_reuse_0									
impl_1_route_report_clock_utilization_0								7/11/26 8:52 PM	6.9 KB
impl_1_route_report_bus_skew_0								7/11/26 8:52 PM	1.1 KB
impl_1_route_implementation_log_0								7/11/26 9:01 PM	23.1 KB
Post-Route Phys Opt Design (post_route_phys_opt_design)									
impl_1_post_route_phys_opt_report_timing_summary_0									
impl_1_post_route_phys_opt_report_bus_skew_0									
Write Bitstream (write_bitstream)									
impl_1_bitstream_report_webtalk_0								7/11/26 9:01 PM	21.1 KB
impl_1_bitstream_implementation_log_0								7/11/26 9:01 PM	23.1 KB

Report						Options		Modified	Size
Synth Design (synth_design)									
synth_1_synth_report_utilization_0								7/11/26 8:50 PM	6.5 KB
synth_1_synth_synthesis_report_0								7/11/26 8:50 PM	26.4 KB
Implementation									
impl_1									
Design Initialization (init_design)									
impl_1_init_report_timing_summary_0									
Opt Design (opt_design)									
impl_1_opt_report_drc_0								7/11/26 8:51 PM	2.5 KB
impl_1_opt_report_timing_summary_0									
Power Opt Design (power_opt_design)									
impl_1_power_opt_report_timing_summary_0									
Place Design (place_design)									
impl_1_place_report_io_0								7/11/26 8:52 PM	96.7 KB
impl_1_place_report_utilization_0								7/11/26 8:52 PM	7.4 KB
impl_1_place_report_control_sets_0								7/11/26 8:52 PM	2.7 KB
impl_1_place_report_incremental_reuse_0									
impl_1_place_report_incremental_reuse_1									
impl_1_place_report_timing_summary_0									
Post-Place Power Opt Design (post_place_power_opt_design)									
impl_1_post_place_power_opt_report_timing_summary_0									
Post-Place Phys Opt Design (phys_opt_design)									
impl_1_phys_opt_report_timing_summary_0									
Route Design (route_design)									

Synthesis

Project Summary

Device

FSM_Traffic_Light_Controller.v

tb_Traffic_Light_Controller.v

Traffic_Light_Controller.xdc

Top_Traffic_Light_Controller.v

Clock_Divid

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⚙️

Tcl Console

Messages

Log

Reports

Design Runs

Timing